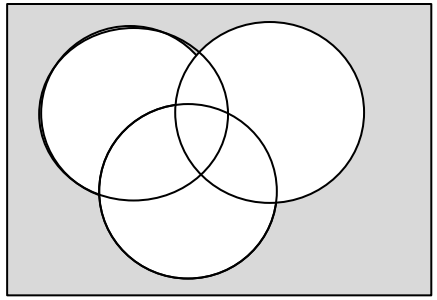
Question	Answer	Marks	Partial Marks	Guidance
18(a)	$\frac{AB}{18} = \cos 60$	**M1**		STM consider knowing the side adjacent to a 60° angle is half the hypotenuse
	$\cos 60 = \frac{1}{2}$ and $AB = 9$	**A1**		
	$[CD^2 =]\ 6^2 + (17 - theirAB)^2$	**M1**		STM consider 3, 4, 5 triangle approach
	$36 + 64$	**B1**		
	$\sqrt{100} = 10$	**A1**	Dep on M1A1M1B1	Accept $CD^2 = 100$ then 10
18(b)	$39 + 9\sqrt{3}$ final answer	**4**	**B3** for $[BE =]\ 9\sqrt{3}$ OR **M1** for $\frac{BE}{18} = \sin 60$ or better **B1** for $\sin 60 = \frac{\sqrt{3}}{2}$	
19		**1**		
20(a)	$14\sqrt{3}$ final answer	**2**	**B1** for $10\sqrt{3}$ or $4\sqrt{3}$	

Question	Answer	Marks	Partial Marks	Guidance
20(b)	$3(\sqrt{7}-2)$ oe final answer	**3**	**B2** for $\frac{9(2+\sqrt{7})}{4-7}$ or better or **M1** for $\frac{9}{2+\sqrt{7}} \times \frac{2-\sqrt{7}}{2-\sqrt{7}}$ oe	Accept e.g. $-3(2-\sqrt{7})$
21(a)	(0, –3)	**1**		
21(b)	[y =] $-\frac{1}{2}x+12\frac{1}{2}$ final answer	**5**	**M1** for [grad AB =] $\frac{15-7}{5-1}$ oe **M1** for –1 divided by *their* grad of AB **B1** for [midpoint =] (3, 11) **M1** for substituting *their* (3, 11) into y = (*their m*)x + c	Accept –0.5 and 12.5 or $\frac{25}{2}$ Correct answer in different form or unsimplified implies 4 marks *their m* ≠ *their* gradient AB e.g. $y - 11 = -\frac{1}{2}(x-3)$
22(a)	(–1, 1)	**5**	**B2** for $3x^2+2x-1$ or **B1** for two terms correct **M1** for setting *their* dy/dx =0 **M1** for $(3x-1)(x+1)$ FT *their* 3 term quadratic **A1** for $x = -1$ [and $x = \frac{1}{3}$]	For B marks isw e.g. 2nd derivative found Accept e.g. formula, complete the square